Martha Fey

Cedar Rapids, Iowa | marthafey17@gmail.com | linkedin.com/in/martha-fey

Research interests: Environmental health, statistical analysis, water quality, environmental epidemiology, and human health outcomes

EDUCATION

Bachelor of Science in Sustainability Science
University of Iowa | Iowa City, Iowa | Aug 2021 – May 2025

GPA: 3.90/4.0 | **Academic standing:** Top 5% of graduating class

RESEARCH EXPERIENCE

Research Assistant
Ecoclimatology of Carbon and Water (ECH2O) Lab, University of Iowa | Mar 2023 – Jun 2025

- Independently constructed an Eddy Covariance Flux Tower to collect continuous ecosystem–atmosphere exchange data.
- Integrated solar power, battery storage, and remote data transmission systems for autonomous environmental monitoring.
- Collected and analyzed environmental datasets to investigate ecosystem processes and climate-related dynamics.

ENVIRONMENTAL HEALTH EXPERIENCE

Environmental Health and Land Use Assistant
Benton County | Vinton, Iowa | Mar 2026 – Present

- Collected water samples from more than 100 private wells and explained test results and recommendations to homeowners.
- Cleaned and organized water-quality data for analysis in R and developed a proposed research approach examining relationships between well characteristics, including age and depth, and water-quality results.

PUBLICATIONS

Pacer, R., Meerdink, S., Swanson, B., Ramker, N., Fey, M., et al. (2026). *Assessing the Biogeophysical Cooling Potential of Constructed Prairie with Eddy Covariance & Thermal UAS Remote Sensing.* Manuscript under review.

TECHNICAL SKILLS

Software: ArcGIS Pro, R, Python

Applied ArcGIS Pro to develop spatial datasets and maps supporting land-use evaluation processes.

AWARDS

Outstanding Fourth-Year Student Award
University of Iowa Department of Geographical and Sustainability Sciences | Mar 2026

Recognized for academic excellence and dedication to learning; received a $1,000 scholarship through the Martha L. Corry Fund for Geography.

IdeaStorm Spring 2024 Competition | 2nd Place
University of Iowa John Pappajohn Entrepreneurial Center

Awarded $400 for developing and pitching a food sensitivity app at the University of Iowa’s IdeaStorm competition.